

Chapter 5

Tribological Implication of Particles

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5.1 INTRODUCTION

When two solid surfaces are brought into contact and are followed by separation or sliding, wear particles are generated at the contact interface, and they behave as contaminants for surfaces and surroundings.

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These particles are generated by adhesive, abrasive, plastic flow, fatigue, corrosive, diffusive, and melting wear mechanisms, some of which take place at the same time in the contact region.^{1,2}

The size of a wear particle ranges from the scale of nanometers to that of micrometers, and the microstructure of the particle material is very different from that of the original surface material in general. Both dry and wet contact systems generate wear particles of this character. Wear particles and the mechanisms of their generation are described in this chapter from this viewpoint.

5.2 MICRO-SITE FOR GENERATION OF WEAR PARTICLES

No solid surface is ideally smooth from a microscopic viewpoint. The cleaved surface of mica sheet is rough at the scale of a silicon atom. The well-polished silicon wafer surface of a single crystalline structure is still rough at the nanometer scale. The surfaces of standard glass mirrors, optical lenses, and fine diamond tips of manipulators have surface asperities whose height ranges from about 10 to 100 nm. The surfaces of machine elements and devices made of metals, ceramics, or plastics have surface asperities whose heights range from about 0.1 to 100 μm .

Figure 5.1 shows an example of surface asperities observed with an AFM (atomic force microscope) on the mirror surface of a silicon wafer. A cross-section of a flat surface as shown in Figure 5.1 is schematically shown in Figure 5.2, where asperities are distributed along the center line of $z=0$ at various heights in the direction of the z -axis and the probability density of the asperity height at the level z is given by the function $\phi(z)$.

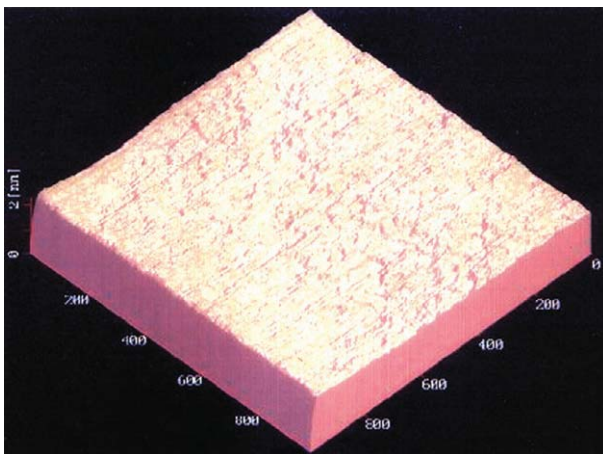


FIGURE 5.1 An AFM image of surface asperities on a polished silicon wafer.

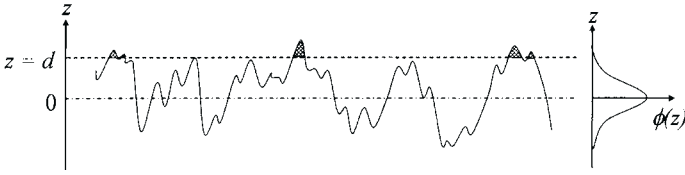


FIGURE 5.2 A schematic model of height distribution of asperities along the center line and the function $\phi(z)$ showing the distribution of probability density of contact of asperities at the level z .

By introducing this function $\phi(z)$ for describing the height distribution of asperities, the number $m(d)$ of asperities in contact against an ideally flat surface at the height of $z = d$ in Figure 5.2 is given by the following equation:

$$m(d) = \eta A_0 \int_d^{\infty} \phi(z) dz \quad (5.1)$$

where η is the density of asperities on the nominally flat surface, A_0 is the nominal surface area, and $\phi(z)$ is generally given by the Gaussian distribution.

The value of η varies depending on the level of surface finish and a certain finishing method does not always give its specific value. Briefly speaking, the value of η may be assumed to vary in the range from 10^2 to 10^4 per mm^2 .

Therefore, when a flat surface has the nominal or apparent surface area of $10 \text{ mm} \times 10 \text{ mm}$ for contact, the total number of asperities for consideration of real contact between asperities is assumed to vary from 10^4 to 10^6 . Among these asperities, the relatively higher asperities contact first and the contact level of $z = d$ shown in Figure 5.2 is determined so as to give the necessary size of the real contact area to support the applied load.

If two asperities are assumed to have a semi-spherical tip shape with radii R_1 and R_2 , a model of frictional contact between them is schematically shown in Figure 5.3, where W is the normal load, F is the friction force for initiating sliding, and A is the real contact area. The mean contact pressure p and the shear strength s_i at the contact interface are defined by

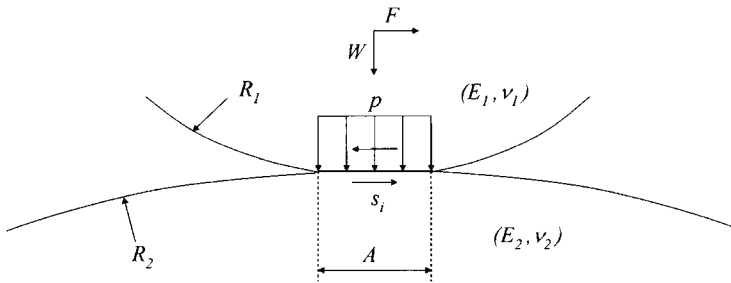


FIGURE 5.3 A schematic model of frictional contact of two spherical asperities. R_1, R_2 , radii of each asperity; E_1, E_2 , Young's moduli; ν_1, ν_2 , Poisson's ratios; W , normal load; F , friction force; A , real contact area; p , mean contact pressure; s_i , shear strength of the contact interface.

$$p = W/A \quad (5.2)$$

$$s_i = F/A \quad (5.3)$$

When the two asperities are in contact with perfectly elastic deformation only, the contact area A is theoretically given by Hertz³ as follows:

$$A = \pi \left(\frac{3}{4} \cdot \frac{WR}{E} \right)^{2/3} \quad (5.4)$$

where $1/R = 1/R_1 + 1/R_2$; $1/E = (1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2$; E_1 , E_2 are Young's moduli; ν_1 , ν_2 are Poisson's ratios.

When either of the asperities is in contact with perfectly plastic deformation only, the contact area A is experimentally confirmed to follow the expression given by McFarlane and Tabor⁴ as follows:

$$A = \frac{W/\sqrt{\alpha k}}{\sqrt{1 - (s_i/k)^2}} \quad (5.5)$$

where α is the experimental constant of 3.3-25, k is the shear strength of the material of the asperity in plastic deformation, and s_i is the shear strength of the contact interface.

By comparing Equations (5.4) and (5.5), it is clear that the contact area A is independent of the shear strength s_i in elastic contact but strongly depends on the value of s_i in plastic contact which is close to zero in ideally lubricated condition and is close to k in high vacuum with uncontaminated surfaces of ductile materials. It is obvious that the smallest value of A is given by $W/\sqrt{\alpha k}$. The value of α changes with the ductility of the asperity material and the shape of contact.⁵

Friction between two asperities given in Figure 5.3 takes place at the contact interface of the area A given by Equations (5.4) or (5.5), and the friction coefficient μ defined as the ratio of the friction force F against the load W is given by Equations (5.2) and (5.3) as follows:

$$\mu = F/W = s_i/p \quad (5.6)$$

From these analyses of micro-mechanisms of contact and friction, it is understood that the contact area of surface asperities determined by Equations (5.4) or (5.5) is the unit base of wear to produce free particles. Micro-contacts due to this mechanism are distributed at the apparent contact interface as shown in Figure 5.2 and generate a large number of wear particles from them in contact and separation or sliding.

5.3 WEAR MODES AND PARTICLES

5.3.1 Adhesive Transfer of Atoms in Contact and Separation

When two uncontaminated solid surfaces are in contact, surface atoms on both surfaces form atomic bonds as they form in the bulk, and the substrates on both sides of

the contact interface are deformed elastically and/or plastically. In many practical cases, the microstructure of solid has micro-defects such as vacancies, voids, inclusions, dislocations, and grain boundaries. Therefore, the strain in the contact region is not perfectly removed microscopically by unloading even when the apparent strain is within the elastic limit.

In the process of unloading, there is no reason for separation to occur only at the contact interface. The substrate around the contact interface has certain distributions of micro-defects and strains. As the result, fracture in unloading may take place in either the contact interface or subsurface. This means that some atoms are possibly transferred to the mating contact surface. In the practical contact and separation between a clean spherical pin tip of tungsten and a clean flat surface of gold in high vacuum, the transfer of atoms of gold to the surface of tungsten was experimentally observed with a field ion microscope.⁶ In the contact and separation between a tip of nickel and a flat surface of gold, the mutual adhesive transfer of atoms of gold and nickel was shown theoretically by molecular dynamics simulation.⁷

5.3.2 Adhesive Transfer of Flake-Like Particles in Sliding Contact

Macroscopic sliding occurs at the contact interface without fracture in its vicinity when the contact surfaces are well contaminated or lubricated and the shear strength s_i of the contact interface is relatively much smaller than the shear strength k of the contact softer material. Generation of wear particles from the contact surfaces does not happen in such a situation. Relative displacement at the contact interface is expected to take place between atoms or molecules of contaminants or lubricating materials.

However, in practice contaminants or lubricants at the contact interface are partially removed by sliding because of the atomic non-uniformity in the surface layer and the surface roughness.

In such micro-bare regions, atomic bonds between original surfaces are partially formed at the contact interface. The distributions of contact stresses of shear and tension or compression in the substrates are complex at the nanometer scale. Cracks are introduced in the substrate by such stresses, which result in adhesive transfer of particles of bulk material to the mating surface.

Figure 5.4 shows three modes of crack initiation and propagation caused by strong adhesion at the contact interface.⁸ In Figure 5.4(a), a fragment is generated as an adhesive wear particle in sliding by having a crack in the lower body at the exit side of the contact and another crack in the upper body at the entry side of the contact. When both cracks reach the front free surfaces, a fragment is formed as a mixture of materials of the upper and lower bodies. It is generally rolled up in the follow-up sliding.⁹ In Figure 5.4(b), a crack is introduced in the lower body at the entry side of the contact and it propagates to the contact interface by generating adhesive transfer of a thin fragment from the lower body to the upper surface. In Figure 5.4(c), a crack is introduced

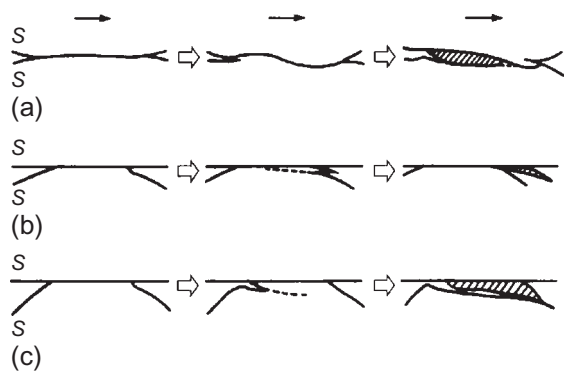


FIGURE 5.4 The modes of crack initiation and propagation for the adhesive transfer of a particle. (a) Curving of the adhesive contact interface → crack initiation → crack propagation on entry and exit sides of contact. (b) Strong adhesion at the contact interface → crack initiation → crack propagation at the exit side of contact. (c) Strong adhesion at the contact interface → crack initiation → crack propagation at the entry side of contact.

in the lower body at the exit side of the contact and it propagates to the free surface on the other side by generating adhesive transfer of a thick fragment from the lower body to the upper surface. These three modes of generation of adhesive transfer of a fragment can take place at any scale from nanometer to micrometer.

Figure 5.5 shows the cluster of thin flake-like fragments which are transferred by the mode of Figure 5.4(b) from the mating surface by adhesion. The thickness of each fragment ranges from about 0.1 to about 1.0 μm .⁸



FIGURE 5.5 Adhesive wear particles of SUS 304 stainless steel transferred from the mating surface by one pass of sliding.⁸

5.3.3 Cutting and Generation of Fine Feather-Like Particles in Abrasive Sliding

When a hard spherical asperity slides on a soft flat surface with a certain amount of indentation, a fine feather-like or ribbon-like particle is generated from the flat surface by a cutting mode. The shear strength s_i of the contact interface must be low enough to generate continuous relative movement at the contact interface without adhesion between contact surfaces. The mechanism of such cutting involves the continuous plastic flow of the soft flat material along the surface of the hard asperity, which is schematically shown in Figure 5.6(a). This mechanism can take place even at the nanometer scale on the surfaces of metals and ceramics. Figure 5.7 shows a cluster of fine feather-like particles of silicon formed by this mechanism at the tip of a diamond pin in one pass of sliding on a silicon wafer.¹⁰

When the shear strength s_i at the contact interface is relatively large because of adhesion at the contact interface and when the amount of indentation of the upper spherical asperity is relatively small, a fine feather-like or ribbon-like particle described in Figure 5.6(a) is not formed, but a wedge-forming mode is generated as described in Figure 5.6(b). Relative movement is taken at the bottom of the wedge in the following process. The wedge is detached from the surface as a wear particle when it meets surface irregularities in the follow-up sliding.

When the indentation of the upper spherical asperity is relatively small and the shear strength s_i is also small, the surface layer of the lower flat surface is slightly deformed and grooved plastically, but a free particle is not generated by a single pass of sliding, as shown in Figure 5.6(c). The accumulation of plastic deformation by the repeated ploughing action is necessary to generate a free

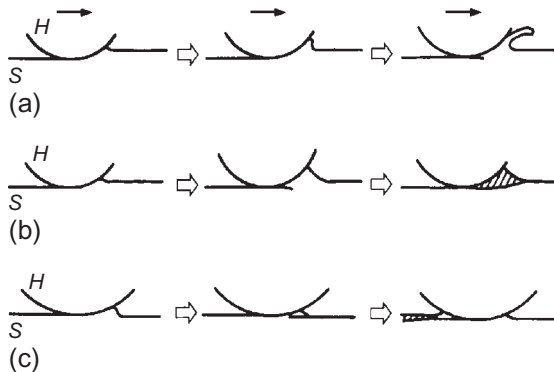


FIGURE 5.6 The modes of plastic deformation and particle generation in abrasive sliding. (a) Fine feather-like or ribbon-like particle generation by cutting. Sliding takes place between the spherical asperity and the particle. (b) Wedge generation. Sliding takes place at the bottom of the wedge as a result of adhesion between the wedge and the spherical asperity. (c) Ploughing with a small amount of plastic deformation on the flat surface without generating a free particle. An accumulated plastic deformation by repeated sliding contact generates a filmy wear particle by the mechanism of plastic flow wear or low-cycle fatigue wear.

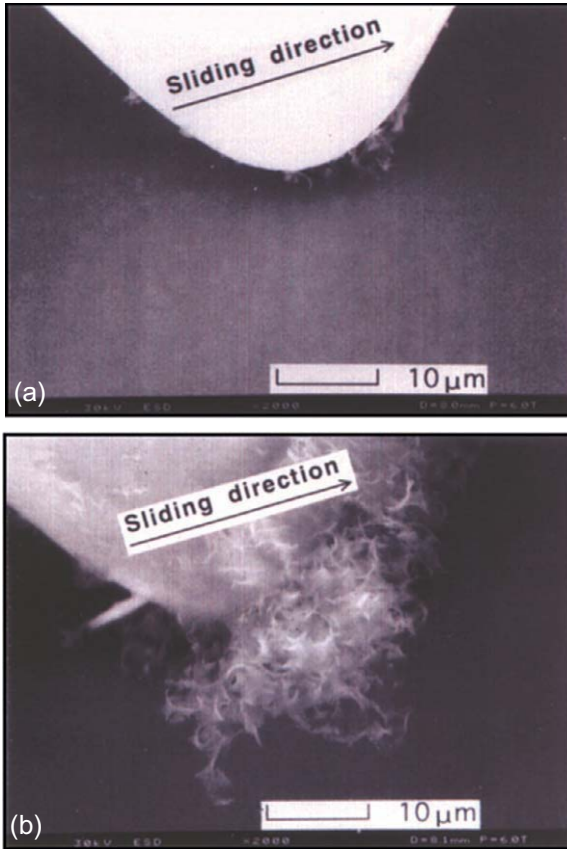


FIGURE 5.7 A cluster of fine feather-like particles of silicon in cutting mode formed by the abrasive sliding of a diamond pin on a silicon wafer after: (a) 2 mm sliding and (b) 20 mm sliding. Small abrasive asperities on the pin tip surface are invisible.¹⁰

particle in ploughing mode from the flat surface by the mechanism of plastic flow or low-cycle fatigue. The regimes of these three modes of wear shown in Figure 5.6 are theoretically described in Figure 5.8 with the degree of penetration D_p and the normalized relative shear strength f defined as follows:

$$D_p = \frac{h(\text{depth of indentation})}{a(\text{radius of contact area})} = R \left(\frac{\pi H_v}{2W} \right)^{1/2} - \left(\frac{\pi H_v}{2W} R^2 - 1 \right)^{1/2} \quad (5.7)$$

$$f = \frac{s_i(\text{shear strength of contact interface})}{k(\text{shear strength of grooved material})} \quad (5.8)$$

where R is the radius of spherical asperity tip, W is the normal load, and H_v is the hardness of the grooved material. This is experimentally confirmed in Figure 5.8, which shows three wear modes of cutting, wedge forming, and ploughing very well.¹¹

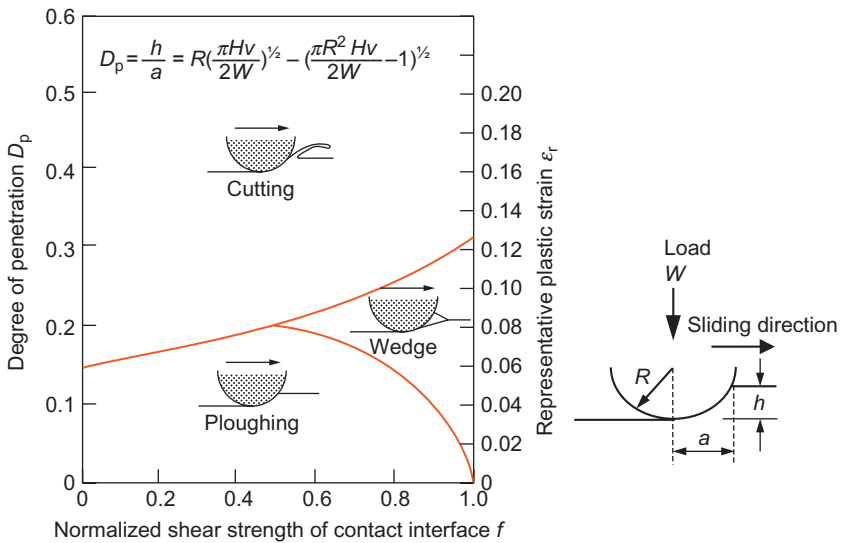


FIGURE 5.8 Abrasive wear mode diagram described by the degree of penetration D_p and normalized shear strength f . W , load; R , radius of asperity; h , depth of indentation; a , radius of contact area; H_v , hardness of grooved material; ϵ_r , representative plastic strain corresponding to D_p .¹¹

5.3.4 Surface Plastic Flow and Thin Film Wear Particle Generation by Repeated Sliding Contact

When a ductile asperity is repeatedly or continuously rubbed by a relatively harder flat surface in plastic contact with low shear strength s_i at the contact interface, the surface layer of the asperity flows plastically and generates a thin filmy wear particle which is smeared out of the interface as shown in Figure 5.9(a). Figure 5.10 shows a thin filmy particle generated by this mechanism at a ridge on a steel surface after 8×10^3 friction cycles in oil.¹²

A similar thin filmy wear particle is generated on a ductile flat surface under the contact condition shown in Figure 5.6(c). In the case of Figure 5.9(b), thin filmy wear particles are smeared out from the surface layer of ductile asperity to the direction vertical to the sliding direction by the repeated or continuous unidirectional shear under compression. The symbol $\otimes$ in the figure denotes the sliding direction of the upper flat surface which is perpendicular to the paper surface.

5.3.5 Crack Initiation and Propagation in the Subsurface of the Contact Region and Generation of a Flake-Like Particle by Repeated Sliding

When an asperity repeatedly slides over the same surface in elastic-plastic contact, a crack is generated in the local region of plastic yield and it propagates within the subsurface until it reaches the surface and generates a thin flake-like particle

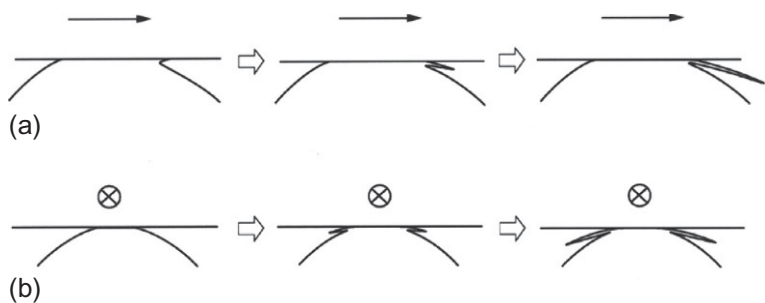


FIGURE 5.9 The modes of plastic flow and generation of thin filmy wear particles on a ductile asperity in the repeated or continuous sliding against a hard flat surface with a relatively low shear strength s_i at the interface. (a) A thin filmy wear particle is smeared out from the surface asperities to the sliding direction by the repeated or continuous unidirectional shear under compression. (b) Thin filmy wear particles are squeezed out from the surface layer to the direction vertical to the sliding direction by the repeated or continuous unidirectional shear under compression. The symbol $\otimes$ denotes the sliding direction of the upper flat surface which is perpendicular to the paper surface.

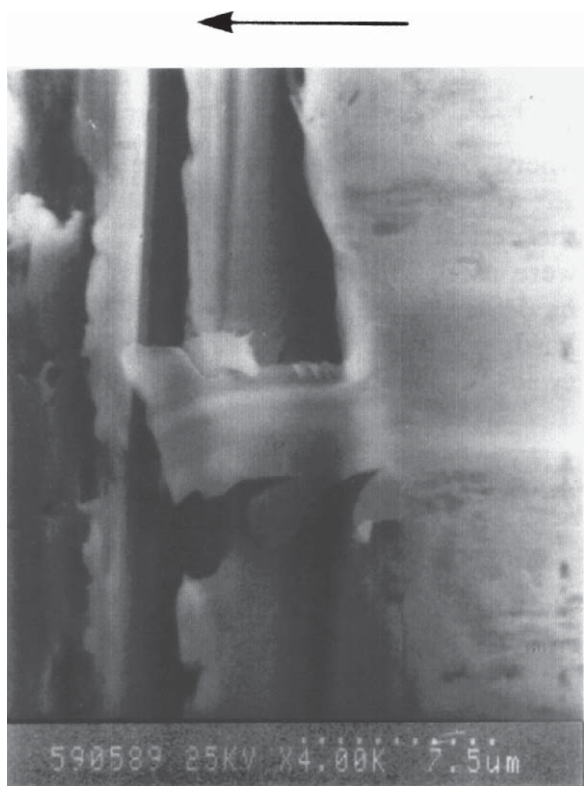


FIGURE 5.10 An SEM image of a thin filmy wear particle of steel smeared out from a contact ridge after 8×10^3 cycles of sliding contact against a steel flat in oil.¹² The arrow indicates the sliding direction of the counter surface in repeated sliding contact.

by the mechanism of fatigue, as shown in Figure 5.11. In Figure 5.12, a very thin flake-like particle is delaminated by the low-cycle fatigue mechanism from the surface of CN_x coating, which has hardness around 25 GPa, after repeated sliding of a diamond pin under the light load in the micronewton range.¹³ Wear particles of similar type can be generated from metals, ceramics, and plastics. Depending

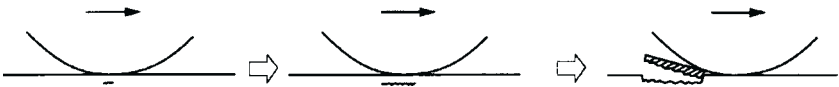
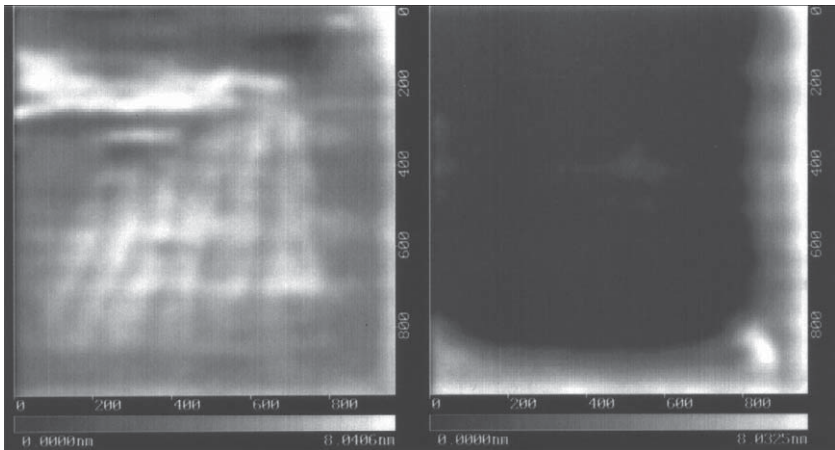
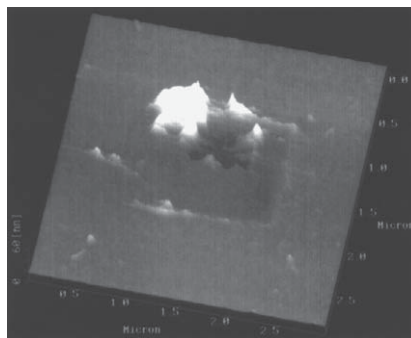


FIGURE 5.11 The mode of crack initiation and propagation for the generation of a flake-like particle by the mechanism of fatigue fracture.



(a) (b)



(c)

FIGURE 5.12 AFM images of delamination of thin flake-like particles from a CN_x coating by the repeated sliding contact of a diamond pin of 15 nm radius under a load of 14 μN .¹³ (a) A wavy pattern of wear scar after 18 cycles of sliding. (b) Sudden surface delamination after 19 cycles. (c) Image of the whole wear scar after 19 cycles.

on the contact geometry and the amount of the contact load, the thickness of flake-like particle ranges from the order of nanometers to micrometers.

5.3.6 Tribo-Oxidation and Generation of Oxide Particles by Repeated Contacts in Air and Water

In the case of sliding contact of steel elements in air without lubrication, contact surfaces are quickly oxidized by friction and the oxide film of iron grows on the rubbed surfaces. The iron oxide itself delaminates from the surface at a certain thickness and forms free particles. The growth of the oxide film is not simply a function of temperature at the interface, which is raised by frictional heat, but is a function of tribological activation of oxidation through deformation and fracture of surface materials and successive mechanical removal of oxides by friction. A wear model based on this understanding of oxidative wear gives an expression for wear coefficient K as follows¹⁴:

$$K = \frac{\text{wear volume/sliding distance}}{\text{load/hardness}} = \delta A \exp(-Q/R_q T) / \xi^2 \rho^2 v \quad (5.9)$$

where ξ is the critical thickness of oxide for self-delamination, δ is the contact length, v is the sliding speed, A is the Arrhenius constant, Q is the activation energy, R_q is the gas constant, T is the temperature, and ρ is the density of oxide.

Tribological activation of oxidation is well observed in rolling contact of silicon nitride Si_3N_4 , which is known as an inert ceramic. Figure 5.13(a) shows fine and transparent fragments of SiO_2 on the surface of a Si_3N_4 roller generated after 10^6 rolling contact cycles between Si_3N_4 rollers in air, and Figure 5.13(b) shows a transparent wear particle of SiO_2 .¹⁵ Figure 5.13 is observed in pure rolling where frictional heat generation is considered negligibly small at the contact interface and the microscopic contact pressure at asperities must

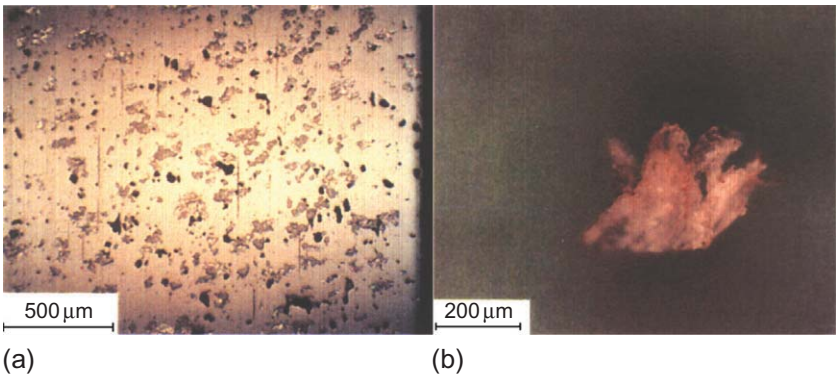


FIGURE 5.13 Optical images of fine and transparent wear particles generated from Si_3N_4 in pure rolling contact in air. Particles are SiO_2 .¹⁵ (a) Transparent particles on the roller surface of Si_3N_4 after 10^6 rolling cycles. (b) A transparent wear particle of SiO_2 .

be close to the hardness value of Si_3N_4 , around 20 GPa. In this contact condition, the oxidation of Si_3N_4 is enhanced at the contact interface in air at room temperature.

When a Si_3N_4 pin slides against a Si_3N_4 disk in water, Si_3N_4 reacts with oxygen and water as follows:



where SiO_2 dissolves in water by forming $\text{Si}(\text{OH})_4$.^{16,17}

When the content of $\text{Si}(\text{OH})_4$ in water is above its saturation point, SiO_2 precipitates as fine particles from the solution. Figure 5.14(a) shows thin films of SiO_2 grown by friction in water at 40 °C. Figure 5.14(b) shows fine needles of SiO_2 formed by sliding friction of Si_3N_4 against itself in water at 20 °C, and Figure 5.14(c) shows fine particles of SiO_2 precipitated from the solution of $\text{Si}(\text{OH})_4$ formed by wear of Si_3N_4 in water.

5.3.7 Wear Particles Generated in Sliding of Steels in Oil with Additives

When steel surfaces slide against each other under boundary lubrication conditions with a base oil (hexadecane), which contains antiwear additives such as zinc dialkyl-dithiophosphate (ZDDP) or meta-tricresyl phosphate (TCP), wear particles of various sizes are generated, as shown in Table 5.1,¹⁹ and Tables 5.2 and 5.3.²⁰

Table 5.1 shows that many wear particles have thickness of about 100 nm and size in the range of 1000–5000 nm. Small wear particles of thickness of about 5–10 nm are also included.

Table 5.2 shows that the typical size of small wear particles (debris) ranges from 10 to 300 nm and the typical size of large particles ranges from 500 to 2000 nm depending on the type of additive. The thickness of wear particle has a range smaller than 500 nm.

Table 5.3 shows that carbon, oxygen, and phosphorus are highly concentrated in the wear particles generated by sliding friction between steels in oil. This means wear particles are made of chemical reaction products of steel, oil, additives, and oxygen.

5.4 WEAR RATE AND NUMBER OF WEAR PARTICLES

The wear rate w_s is generally defined as the wear volume V divided by the load W and the sliding distance L as follows:

$$w_s = V/WL \quad (5.12)$$

It is called specific wear volume or specific wear amount and has the dimensions of mm^3/Nm for convenience of engineering usage.

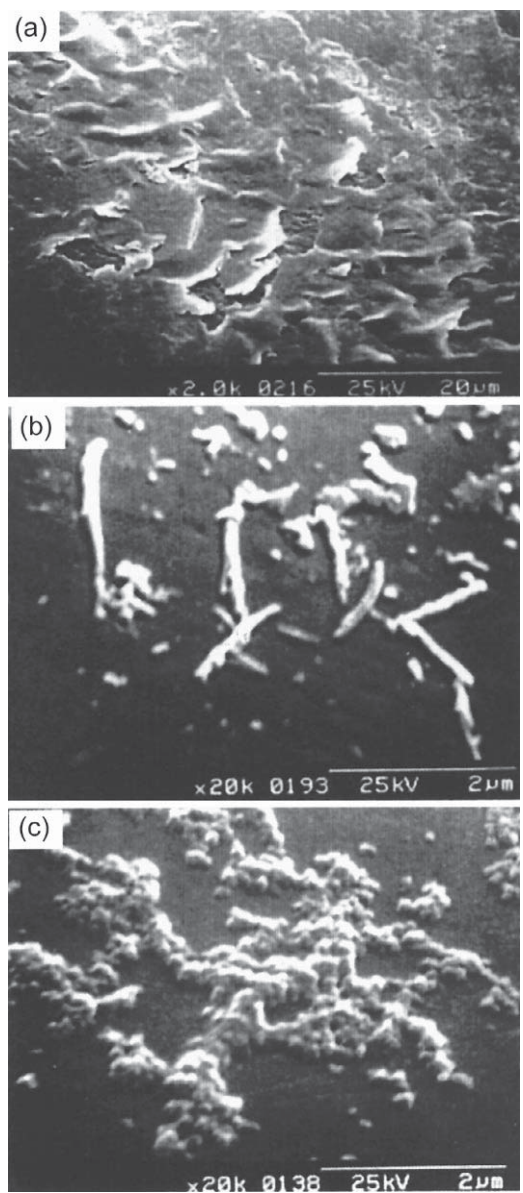


FIGURE 5.14 Wear particles formed on the surface of Si_3N_4 by the self-mated sliding in water.¹⁸
(a) Thin films of SiO_2 on the surface of Si_3N_4 grown at 40 °C. (b) Fine needle-like particles rolled up from SiO_2 film at 20 °C. (c) Fine particles precipitated from water containing $\text{Si}(\text{OH})_4$ generated by wear of Si_3N_4 .

TABLE 5.1 Mean Size of Wear Particles¹⁹

Nature of Interfacial Material	Mean Size (nm)		h/h^*	d/d^*
	h	d		
<i>Without additives</i>				
Delaminated plates	100	1000-5000	1	10^{-2}
Chips	500-1000	$\cong 5000$	1	10^{-2}
Particles	100	100	$\cong 10^{-1}$	10^{-3}
<i>With antiwear additives</i>				
Particles	5-10	3000	1	10^{-5}
Rolls	$\cong 100$	500	1	2×10^{-3}
Lumps	$\cong 100$	> 5000	1	2×10^{-2}
Films	$\cong 100$			2×10^{-3}
<i>h, height; d, lateral mean size; h*, interfacial thickness; d*, Hertzian contact diameter ($\sim 1.5 \times 10^3$ nm).</i>				

TABLE 5.2 Shape and Size of Wear Particles Generated in Boundary Lubrication²⁰

Additive	Film Formed on Scar	Typical Large Debris (μm)	Typical Small Debris (nm)
Second ZDDP	Thick $\sim 0.5 \mu\text{m}$	3×2 (composite)	50
TCP	Thick $\sim 0.2 \mu\text{m}$	0.5 (few, composite)	120
Amine phosphate	Thin ^a	0.5 (composite)	300
Amine monothiophosphate	Thin	0.5 (flakes)	200
S/P	Thin	0.5 (composite)	80
Thiocarbamate	Thin	Gel	30
Base oil	Thin	2 (composite)	10

^aThin refers here to primarily bare surface by optical microscopy, but typically with scattering of thick patches of film.

TABLE 5.3 Chemical Composition of Wear Particles (Relative Units) and Wear Surfaces ²⁰						
Sample	Analysis Technique	Location	C	O	P	Fe
On ball (5 min)	Auger	Scar	11	42	10	34
		Film on scar	6	52	15	24
Gold extraction replica (6.5 min)	Auger	Debris ahead of scar	31	44	8	17
Carbon adhesive replica (6.5 min)	SEM/EDX ^a		63	24	2	1.3
Centrifuge (30 min)	SEM/EDX		86	12	0.7	0.5
Filter (60 min)	SEM/EDX		75	22	1	0.1
^a SEM, scanning electron microscopy; EDX, energy dispersive X-ray.						

When a non-dimensional coefficient is preferred to describe the wear rate, the wear coefficient K is introduced by the following definition:

$$K = \frac{V/L}{W/H_v} \tag{5.13}$$

where H_v is the hardness of wearing material and W/H_v gives an estimate of the real contact area in plastic contact. H_v has the dimensions of N/m^2 , so K is non-dimensional. The value of H_v is of the order of 1 GPa for metallic materials and 10 GPa for ceramics. The value of w_s expressed in units of nm^3/Nm is about 10 times smaller than the value for K .

The value of wear coefficient K for pure metals in self-mated sliding in air varies in the range 10^{-3} to 10^{-2} , as shown in Table 5.4.²¹ The value of K of AISI 52100 Steel in self-mated sliding is of the order of 10^{-2} in dry argon and it is reduced to the order of 10^{-10} in engine oil, as shown in Table 5.5.²²

Figure 5.15 shows the distribution of the specific wear amount w_s and the friction coefficient μ observed with Al_2O_3 , ZrO_2 , SiC , and Si_3N_4 in self-mated sliding in air, where w_s varies from 10^{-9} to $10^{-2} \text{ mm}^3/\text{Nm}$ depending on the load and velocity. Similar distributions of w_s are found with $\text{Si}_3\text{N}_4/\text{Si}_3\text{N}_4$, SiC/SiC , PSZ/PSZ (partially stabilized zirconia), and $\text{Al}_2\text{O}_3/\text{Al}_2\text{O}_3$.²⁴ If the self-mated sliding of Si_3N_4 or SiC is carried out in water, the specific wear amount w_s changes from 10^{-6} to $10^{-8} \text{ mm}^3/\text{Nm}$ for both ceramics by repetitive friction cycles.²⁵

TABLE 5.4 Wear Coefficient (K) in Self-Mated Sliding of Pure Metals in Air²¹

Materials Combination	Wear Coefficient (K)
Silver/silver	4.0×10^{-3}
Cadmium/cadmium	5.7×10^{-3}
Copper/copper	1.1×10^{-2}
Platinum/platinum	1.3×10^{-2}
Zinc/zinc	5.3×10^{-2}

TABLE 5.5 Wear Coefficient (K) in Self-Mated Sliding of AISI 52100 Steel²²

Environment	Wear Coefficient (K)
Dry argon	1.0×10^{-2}
Dry air	1.0×10^{-3}
Cyclohexane	8.4×10^{-6}
Paraffinic oil	3.2×10^{-7}
Paraffinic oil+TCP*	3.3×10^{-9}
Engine oil	2.0×10^{-10}

*TCP: Meta-tricresylphosphate

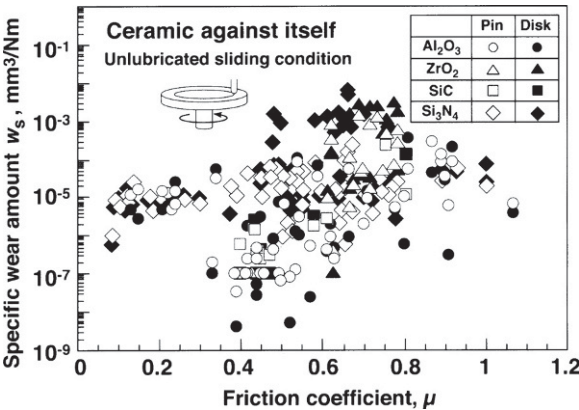


FIGURE 5.15 The specific wear amount w_s and friction coefficient μ observed by self-mated sliding of Al₂O₃, ZrO₂, SiC, and Si₃N₄ in air without lubrication.²³

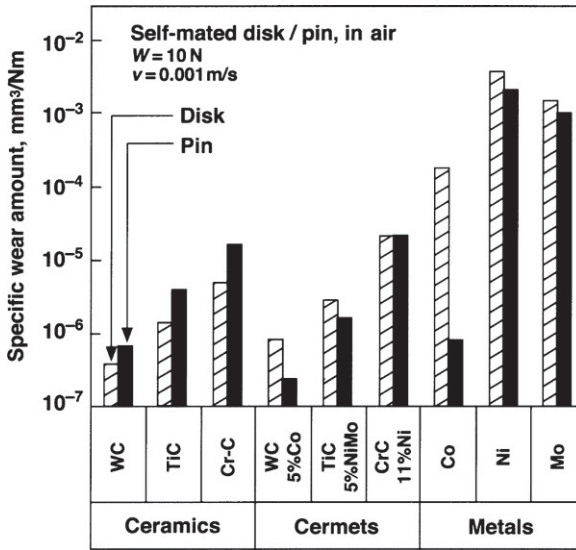


FIGURE 5.16 The specific wear amount w_s of ceramics, cermets, and metals, where the ceramics are used as such and the metals are used as binders for cermets. Observations are made under the same load and velocity in air as for self-mated materials.²³

Figure 5.16 shows a comparison of the specific wear amount w_s of ceramics, cermets, and metals, where w_s of ceramics and cermets varies in the range from 10^{-7} to $10^{-5} \text{ mm}^3/\text{Nm}$ and that of the metals varies in the range from 10^{-4} to $10^{-3} \text{ mm}^3/\text{Nm}$.²⁶

By considering all these results shown in Tables 5.4 and 5.5 and in Figures 5.15 and 5.16, the value of w_s can be assumed to vary in a wide range from 10^{-10} to $10^{-2} \text{ mm}^3/\text{Nm}$, depending on the contact material, load, velocity, environment, and lubricant. Only when lubrication is very well applied and the contact surfaces are well separated by a lubricant film, a specific wear amount smaller than $10^{-10} \text{ mm}^3/\text{Nm}$ can be expected.

If we assume that all wear particles have the same size and shape of a thin rectangular plate with thickness t , width l , and length l , the volume ΔV of a wear particle is described by

$$\Delta V = tl^2 \quad (5.14)$$

By introducing the number N of total wear particles for the wear volume V , V is described by

$$V = Ntl^2 \quad (5.15)$$

It is generally observed from experiments that $t/l \approx 1/10$. By introducing this experimental relationship into Equation (5.15), the volume V becomes

$$V = \frac{N}{10} l^3 \quad (5.16)$$

By introducing Equation (5.16) into Equation (5.12), the number N of total wear particles under load W and after sliding distance L is described by

$$N = \frac{10}{l^3} w_s WL \quad (5.17)$$

If we assume the value of $l = 10^{-4}$ mm and $w_s = 10^{-6}$ mm³/Nm, Equation (5.17) gives

$$\frac{N}{WL} = 10^7 \text{ (1/Nm)} \quad (5.18)$$

The value given by the brief calculation above shows that a very large number of wear particles are generated even when the value of w_s is acceptably small in practice. If a man of 1000 N weight walks with a slip of 1 mm per step, 10^7 wear particles will be generated by his one step.

For more exact prediction of the number of wear particles generated at a certain wear rate, the distributions of particle size and shape must be more precisely determined. The number of total wear particles generated under unit load and unit sliding distance must be much larger than the value given by Equation (5.18) if wear particles in the size range below 100 nm are counted in the calculation.

5.5 SIZE AND NUMBER OF WEAR PARTICLES BY SLIDING

Figure 5.17 shows the friction apparatus inversely set in the scanning electron microscope (SEM), where a rotary disk of 18Cr-8Ni stainless steel slides

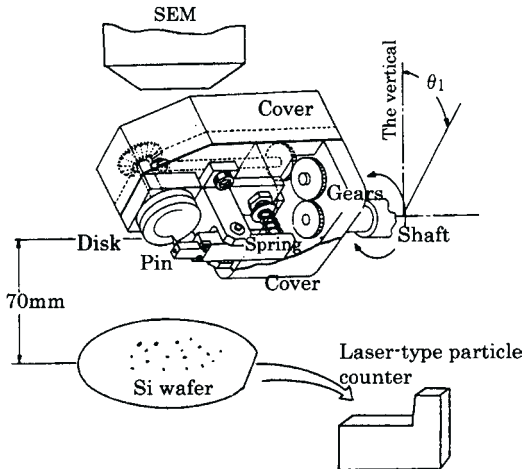


FIGURE 5.17 Friction apparatus of pin-on-disk which is inversely set in the SEM and has a silicon wafer 12.7 cm (5 in.) in diameter for collecting wear particles falling from the contact by gravity in a vacuum of 9×10^{-3} Pa.²⁷

against a hard pin and wear particles of stainless steel fall on the Si wafer in a vacuum of 9×10^{-3} Pa.²⁷ The size and number of wear particles on the Si wafer are measured by the laser-type particle counter, where the size is given by the radius of circular area equivalent to the area of the wear particle. The wear mode is confirmed with the SEM image of the contact region which is observed by reversing the friction apparatus to the normal position.

Figure 5.18 shows the distribution of wear particle size δ described with its probability density $\phi(\delta)$ observed for the WC pin/stainless steel disk in 10 friction cycles of 1 m sliding distance.²⁸ The pin tip radius, contact load, and number of wear particles are also shown in the figure. The size δ is measured in the range 0.25–10 μm . It is obvious from the figure that $\phi(\delta)$ rapidly increases from 1 to 14 μm^{-1} as δ decreases from 0.40 to 0.25 μm , and it decreases relatively slowly from 0.15 to below 0.02 μm^{-1} as δ increases from 1 to 10 μm . The observed wear mode for the results in the figure is wedge forming shown in Figures 5.6(b) and 5.8, which means that the wear particle size in a wear mode is distributed over a wide range from nanometers to micrometers in a complex way. The mean value of δ for the total wear particles observed in Figure 5.18 is about 2 μm .²⁸

The effects of pin tip radius R (μm), contact load (N), and pin material on the number of wear particles n (m^{-1}) in sliding of 18Cr-8Ni stainless steel disk against diamond and WC pins are shown in Figure 5.19.²⁹ The dominant wear

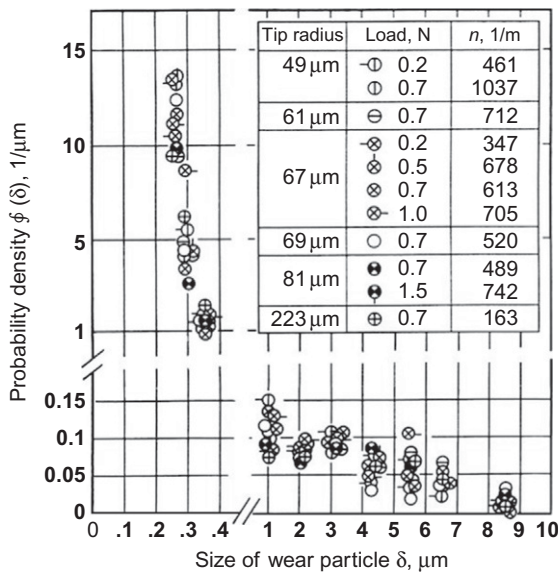


FIGURE 5.18 The distribution of wear particle size δ described in terms of its probability density $\phi(\delta)$ observed for the WC pin/stainless steel disk in 10 friction cycles of 1 m sliding distance. The used pin tip radius, contact load, and the number of wear particles n are shown together.²⁸

mode of stainless steel observed for the diamond pin is the ploughing shown in Figures 5.6(c) and 5.8 under a contact load from 0.2 to 0.7 N. The cutting wear mode shown in Figures 5.6(a) and 5.8 is only observed under the very severe contact conditions of 1 N load and 29 μm tip radius.

The dominant wear mode of stainless steel observed for the WC pin is wedge forming under 0.7 N load for the whole range of pin tip radius from 49 to 223 μm . It is obvious from Figure 5.19 that the number of wear particles n (m^{-1}) from the stainless steel disk is very sensitive to the change of pin tip radius, contact load, or pin material.

In order to see the combined effect of pin tip radius R and contact load W on the number of wear particles n in Figure 5.19, the degree of penetration D_p defined by Equation (5.7) is introduced as an index including parameters R and W , and Figure 5.20 is described with n and D_p for the sliding of 18Cr-8Ni

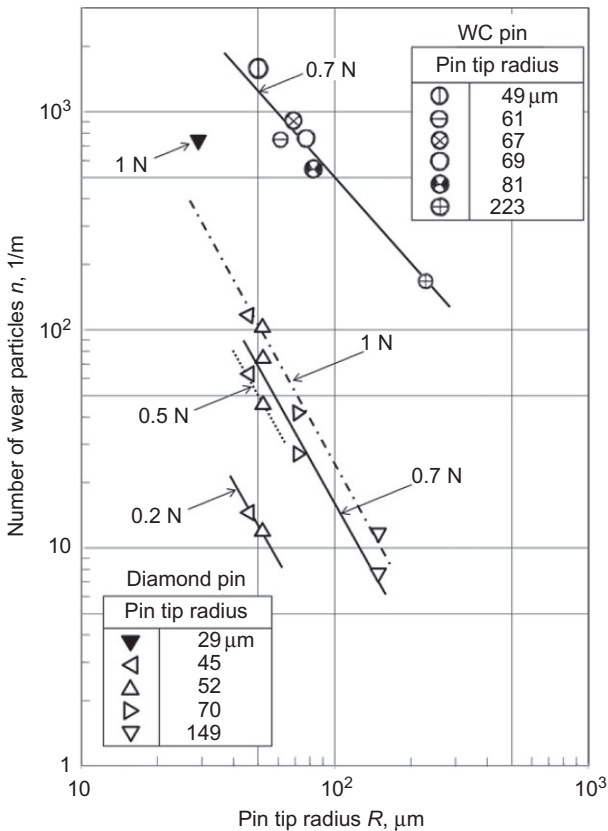


FIGURE 5.19 The effects of pin tip radius R (μm), contact load (N), and pin material on the number of wear particles n (m^{-1}) in sliding of 18Cr-8Ni stainless steel disk against pins of diamond and WC.²⁹

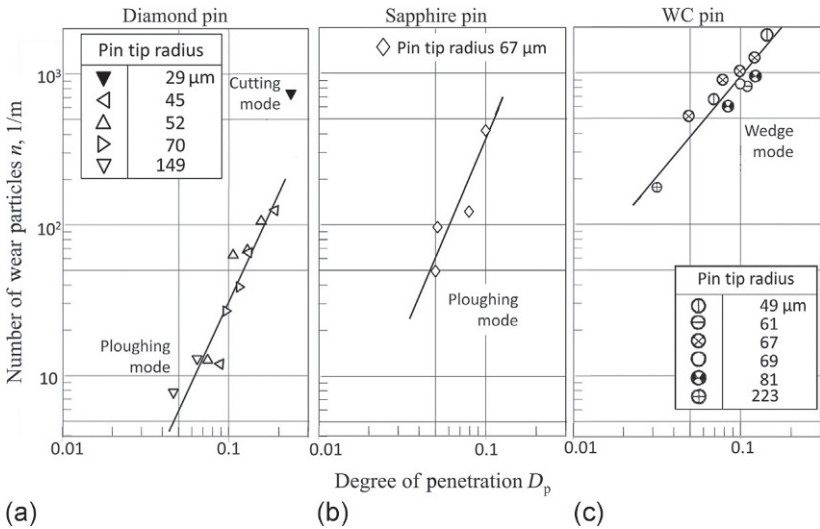


FIGURE 5.20 Relationship between the number of wear particles n (m^{-1}) and degree of penetration D_p for the 18Cr-8Ni stainless steel disk sliding against (a) diamond pin, (b) sapphire pin, and (c) WC pin.²⁹

stainless steel disk against (a) diamond pin in ploughing wear mode, (b) sapphire pin in ploughing wear mode, and (c) WC pin in wedge-forming wear mode.

The experimental relationships between n and D_p in the diagrams of Figure 5.20(a)–(c) are obtained as follows:

- For diamond pin/stainless steel disk

$$n = 6.0 \times 10^3 D_p^{2.3} \quad (5.19)$$

- For sapphire pin/stainless steel disk

$$n = 9.8 \times 10^4 D_p^{2.3} \quad (5.20)$$

- For WC pin/stainless steel disk

$$n = 1.1 \times 10^4 D_p^{1.2} \quad (5.21)$$

It is clear from Figure 5.20 and Equations (5.19)–(5.21) that the diamond pin generates the smallest value of n , the sapphire pin an intermediate value of n , and the WC pin the largest value of n for the same value of D_p . Briefly speaking, the diamond and sapphire pins generate smaller numbers of steel wear particles with the same power law $n \propto D_p^{2.3}$ in ploughing wear mode, and the WC pin generates larger number of steel wear particles with a different power law $n \propto D_p^{1.2}$ in wedge-forming wear mode.

The wear mode of disk material at a certain value of D_p is determined by the value of normalized shear strength f defined by Equation (5.8) at the contact interface between the pin and disk, as shown in Figure 5.8. For example, when $D_p = 0.1$ the ploughing wear mode is expected to occur as shown in the figure for f values below 0.85 and the wedge-forming wear mode for f values above 0.85. The effect of pin material on parameter n is understood from the effect of f on wear mode in the abrasive wear mode diagram shown in Figure 5.8.

5.6 CONCLUDING REMARKS

5.6.1 Solid Wear Particles

The representative wear mechanisms of solids (metals, ceramics, and hard coatings) and characteristics of wear particles are explained in this chapter by introducing wear models and experimental observations of wear in vacuum, air, water, and oil. For the microscopic understanding of the wear mechanism, the mechanisms of contact and friction are briefly introduced first.

The mechanisms of generation of wear particles introduced in this chapter are applicable for particles in the size range from nanometers to micrometers. Experimental observations of wear particles in the range of 1-10 nm are lacking. However, the wear models valid in the range of micrometers can also be applied to wear at the nanometer scale, because the usefulness of the traditional contact mechanics based on the continuum theory is very well confirmed for explaining the results by molecular dynamics simulations.^{30,31}

5.6.2 Generation of Gas Molecules by Wear

In the unlubricated sliding of diamond against hard materials in air, a high temperature is generated in the contact region as a result of frictional heating, and diamond is worn by oxidation generating carbon dioxide (CO_2).³² If silicon nitride (Si_3N_4) is rubbed in air, it is oxidized and NH_3 gas is generated together with silicon dioxide (SiO_2) as shown in Equation (5.10).^{17,18} The gases CO_2 and NH_3 are contaminants at the molecular scale for various clean surfaces. These kinds of gas molecules generated by wear are not explained in this chapter, because published reports are very few and are insufficient to provide useful information from the viewpoint of contamination control.

5.6.3 Triboemission of Electrons, Ions, Photons, and Particles

Electrons, ions, photons, and particles, which are charged negatively or positively, are emitted from the contact region during friction.³³ The extent of their emission increases with load and sliding velocity. Tribo-microplasma is generated near the contact region and it introduces unique tribo-chemical reactions.^{33,34} Figure 5.21 shows (a) total, (b) UV (ultraviolet), and (c) IR (infrared) images

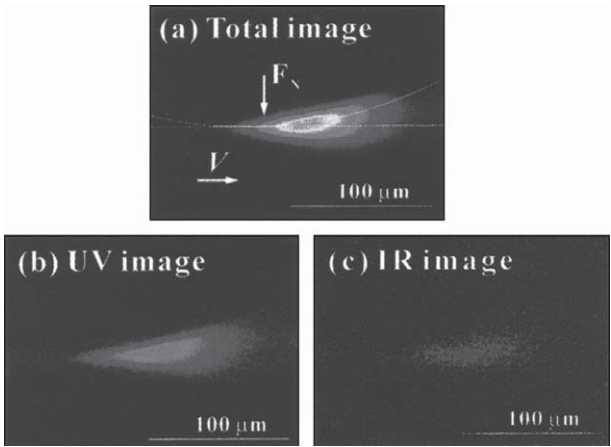


FIGURE 5.21 The images of tribo-microplasma observed from the sliding contact side between a spherical diamond pin and a sapphire disk.³⁵ (a) Total side image. The arrow for V denotes the sliding direction of the lower disk. The arrow for F denotes the loading direction at the center of contact. (b) UV (ultraviolet) image. (c) IR (infrared) image.

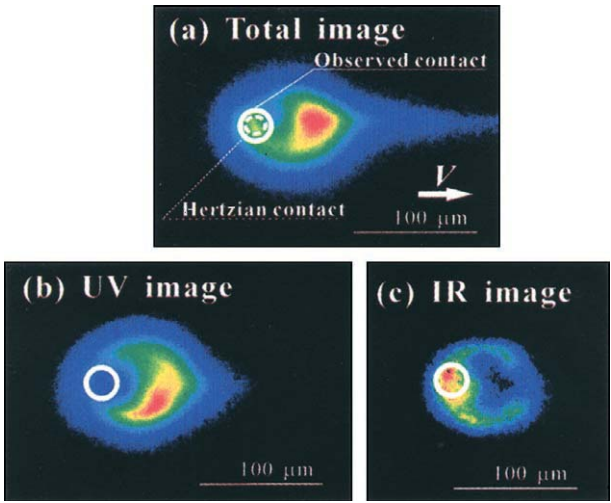


FIGURE 5.22 The images of tribo-microplasma observed from the bottom of sliding contact between a spherical diamond pin and a sapphire disk. The open circles in the images denote the observed contact areas.³⁵ (a) Total image. The arrow for V denotes the sliding direction of the lower disk. (b) UV image. (c) IR image.

of a tribo-microplasma observed from the sliding contact side between a spherical diamond pin and a sapphire disk.³⁵ Figure 5.22 shows the (a) total, (b) UV, and (c) IR images of the same plasma observed from the bottom of contact through the sapphire disk.³⁵ The plasma observed in these figures is thought to enhance chemical reactions around the contact region, and the resultant chemical reaction

products may act as contaminants in some cases. However, a practically useful explanation of the plasma effect on contamination is not discussed in this chapter, because this tribo-microplasma was found only recently^{35,36} and further research is necessary from the viewpoint of contamination control.

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